

MB 519: Creative Thinking, Problem Solving and Decision Making

Course Code	MB 519
Course Name	Creative Thinking, Problem Solving and Decision Making
L-T-P-C	2-0-0-2 (L-T-P-C)
Prerequisites	None
Intended for	MBA
Distribution	Compulsory
Approval	50th BoA (Jul 2023)

Course Contents

- **Introduction:** Understanding problem solving and decision making, Types of problems and decision making. (2 Hours)
- **Human Heritage for Problem Solving and Decision Making:** Creativity and inspiration, Intuition, knowledge, intelligence, wisdom and creativity, empathy, Creativity and evolution. Thinking and its Types, Lean thinking, critical thinking, lateral thinking and design thinking methods, Divergent and convergent thinking. (5 Hours)
- **Reasoning with Data:** Types of reasoning, deductive vs. inductive reasoning, reasoning with data, role of assumptions and biases, evaluating assumptions, biases in inductive reasoning for handling data, avoiding deductive and inductive reasoning fallacies, abductive reasoning, abduction in the field of artificial intelligence, reasoning backwards; Logical, probabilistic and geometric reasoning, problem solving and thinking traps, and their avoidance. (5 Hours)
- **Analytical thinking and decision making:** Analytical Thinking and decision making, stages of analytical thinking, data analytic thinking, Analytic hierarchy and network process. (3 Hours)
- **Problem framing and solution:** Identifying and defining problems, building a model, solving the problem through pattern finding, simplifying and eliminating, developing alternatives and evaluating options, what-if analysis, Complex problem solving, six thinking hats. (4 Hours)
- **Tools, Techniques and Skills:** Interpretation and Ideation techniques: Empathy mapping, mind mapping, journey maps, affinity and cause-effect diagram, pattern recognition; Brainstorming, brain dump, value proposition canvas, SCAMPER. Analysis Methods: Paired Comparison Analysis, Six Thinking Hats, Cost/Benefit Analysis, Decision Trees, Pareto Analysis, Grid Analysis, PMI, Force Field Analysis, Root-Cause Analysis, storyboarding. (5 Hours)
- **Simulation and Optimization for Problem solving and decision making:** Simulation and its types, importance of simulation and optimization for problem solving and decision making, role of data and model in simulations. physical simulation vs. computer simulation. (4 Hours)

Lab Exercises (If applicable)

Lab to be conducted on a 2-hour slot. It will be conducted in tandem with the theory course so the topics for problems given in the lab are already initiated in the theory class. The topics taught in the theory course should appropriately be sequenced for synchronization with the laboratory.

References

1. T.H. Davenport and J. Kim, **Keeping up with the Quants**, Harvard Business Review Press, Boston, MA (rs. 1,555), 2013.
2. Daniel Kahneman, **Thinking, Fast and Slow**, Penguin Random House, 2012.
3. Robert J. Sternberg Ed., **Thinking and Problem Solving**, 2nd Edition, Academic Press, 1994.
4. de Bono, E., **Six Thinking Hats**, MICA Management Resources, 1999.
5. Eugene O'Loughlin, **An Introduction to Business System Analysis**, The Liffey Press, Ireland, 2009.
6. Gerald F. Smith, **Quality Problem Solving**, ASQ Quality Press, 1998.
7. Wisconsin G. Polya, **How to Solve It**, Princeton University Press, 1988
8. Edward B. Burger and Michael Starbird, **5 Elements of Effective Thinking**, Princeton University Press, Oxfordshire, 2021.
9. Jeanne Liedtka, Andrew King and Kevin Bennett, **Solving Problems with Design Thinking**, Columbia University Press, 2013.
10. Luc De Brabandere and Alan Iny, **Thinking in New Boxes**, Random House, 2013.
11. James L. Adams, **Conceptual Blockbusting**, Basic Books, 2019.
12. Tom Kelley and D. Kelley, **Creative Confidence**, William Collins, 2013.
13. J. Butterfield, **Problem Solving and Decision Making**, Cengage Learning, 2010.
14. Saaty, T.L., **Creative Thinking Problem Solving and Decision Making**, RWS Publications, 2008.
15. Paulos, J.A., **Innumeracy: Mathematical Illiteracy and Its Consequences**, New York: Hill and Wang.
16. Fisher, A., **Critical Thinking: An Introduction**, Cambridge University Press, 2001.